

Ramification of Knots

MA461 Algebraic Topology

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What is a knot?

A knot is an embedding of the circle S^1 into a 3-manifold M .

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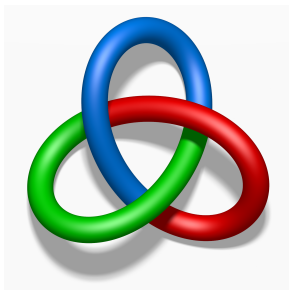


Figure: Trefoil knot

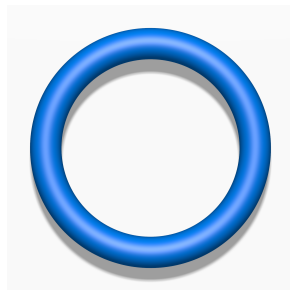


Figure: Trivial knot

Tubular Neighborhoods

A *normal tube* V_C of a curve C is constructed as follows:

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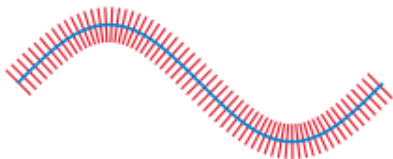


Figure: Tubular neighborhood of a curve

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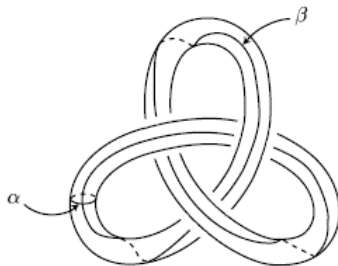


Figure: Trefoil knot with meridian and longitude

The Knot Group

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$M - V_K$ is a deformation retract of $M - K$, so
 $\pi_1(M - V_K) \cong \pi_1(M - K)$.

Links

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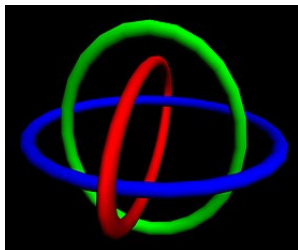


Figure: Borromean rings

Links in S^3

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Theorem: Link groups in S^3

Let $L \subset S^3$ be an r -component link and let V_L be a tubular neighborhood of L . Then $H_1(S^3 - V_L) \cong G_L \cong \mathbf{Z}^r$, where the generators of the group are given by the image of a meridian for each component of the link.

In particular, $H_1(S^3 - V_K) \cong G_K \cong \mathbf{Z}$ for any knot $K \subset S^3$.

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Definition of ramified covering

Let M, N be 3-manifolds; let $f : N \rightarrow M$ be a continuous map and define

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e is called the *ramification index* of f , and $f|_{N-S_N}$ is the covering associated to f .

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In 2-manifolds, the condition is that $(g_e) \circ \varphi = \psi \circ f$, where φ, ψ are homeomorphisms to D^2 . This is saying that f "behaves like" exponentiation to the e -power: $\varphi(z)^e = \psi(f(z))$

Constructing ramified coverings of knots in S^3

Let $K \subset S^3$ be a knot, V_K a tubular neighborhood of K , $X_K := S^3 - V_K$, and $G_K := \pi_1(X_K)$.

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For $n \in \mathbf{N}$, let $\psi_n : G_K \rightarrow \mathbf{Z}/n\mathbf{Z}$ via ψ_∞ and the natural projection $\mathbf{Z} \rightarrow \mathbf{Z}/n\mathbf{Z}$. Let $\rho_n : X_n \rightarrow X_K$ be the covering corresponding to $\ker(\psi_n)$; denote by τ_n the generator of $\text{Gal}(X_n/X_K)$.

Seifert Surfaces

A *Seifert Surface* of a link $L \subset S^3$ is a compact, connected, oriented surface S embedded in S^3 such that $\partial S = L$ and the orientation of L agrees with the induced orientation from S .

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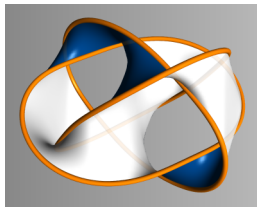


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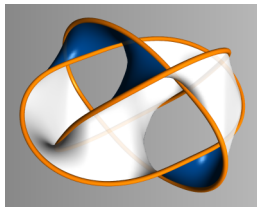


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Theorem: (Frankl and Pontryagin, 1930)

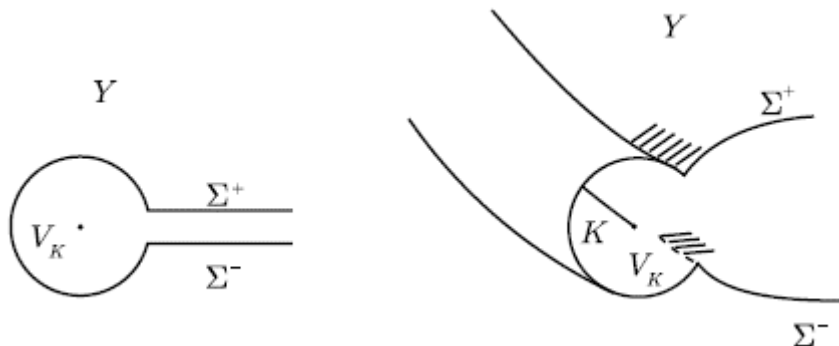
Let L be a link in S^3 . Then there exists a Seifert Surface S of L .

Explicitly constructing X_n

Let Σ_K be a Seifert surface of K ; let Y be the space obtained by cutting X_K along $X_K \cap \Sigma_K$.

Let Σ^+ and

Σ^- be the resulting copies of $\Sigma_K \cap X_K$ in Y as in the following diagrams:



Explicitly constructing X_n

Take $Y_0, Y_1, \dots, Y_{n-1}$ as n copies of Y and form the adjunction by identifying Σ_k^+ with Σ_{k+1}^- (where Σ_{n-1}^+ identifies with Σ_0^-). Let this space be X_n .

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A similar construction gives X_∞ : take Y_k for $k \in \mathbf{Z}$ and identify Σ_k^+ with Σ_{k+1}^- for all k .

Explicitly constructing X_n

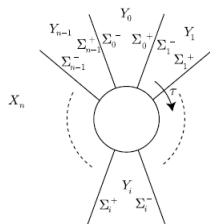


Figure: The covering space X_n

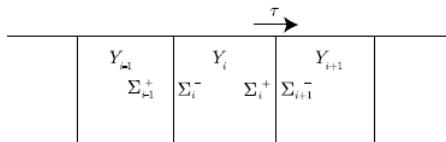


Figure: The covering space X_∞

The Fox Completion

As constructed, X_n and X_K are 3-manifolds with boundary.

We can transform X_n into a closed 3-manifold by adjoining $D^2 \times S^1$ (which is homeomorphic to V_K) to the empty cavity.

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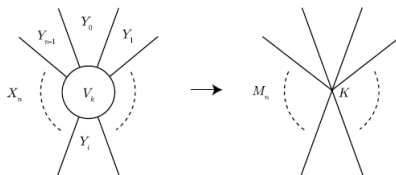


Fig. 2.16

Figure: Fox completion of X_n

Theorems on ramified coverings

Theorem: Fox Completions

Let M be an orientable connected closed 3-manifold and let $L \subset M$ be a link. Let $X := M - L$ and let $h : Y \rightarrow X$ be a finite covering. Then there exists a unique covering $f : N \rightarrow M$ ramified over L such that $h : Y \rightarrow X$ is the associated covering.

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Theorem: (Alexander)

Let M be a connected, oriented 3-manifold. Then there is a finite covering M/S^3 which is ramified over a link $L \subset S^3$

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Setting up

Let $h : M \rightarrow S^3$ be a finite Galois covering of connected, closed, oriented 3-manifolds ramified over a link L . Define $X := S^3 - L$, $Y := M - h^{-1}(L)$, $G := \text{Gal}(Y/X) = \text{Gal}(M/S^3)$ and $n := |G|$.

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Let $K \subset S^3$ be a knot which is disjoint from L or a component of L (such that $K \cup L$ is a link). Suppose $h^{-1}(K) = K_1 \cup \dots \cup K_r$ (an r -component link in M).

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For V_K a tubular neighborhood of K , let V_{K_i} be the connected component of $h^{-1}(V_K)$ such that $K_i \subset V_{K_i}$. Set $X_K := S^3 - V_K$ and fix $x \in \partial X_K$. Finally, suppose $h^{-1}(x) = \{y_1, \dots, y_n\}$.

The decomposition and inertia groups

Since Y/X is a Galois covering, G acts transitively on the set of knots $S_K := \{K_1, \dots, K_r\}$ over K . Define the *Decomposition group* of K_i to be the stabilizer:

$$G_{K_i} := \text{Stab}_G(K_i) := \{g \in G : g(K_i) = K_i\}.$$

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Restriction of g to K_i produces a Galois covering K_i/K and induces a homomorphism $G_{K_i} \rightarrow \text{Gal}(K_i/K)$. The kernel of this morphism is the *Inertia group*:

$$I_{K_i} := \ker(G_{K_i} \rightarrow \text{Gal}(K_i/K)) = \{g \in G_{K_i} : g|_{K_i} = \text{id}_{K_i}\}.$$

Decomposition and Inertia spaces

Since G_{K_i} and I_{K_i} are subgroups of $\text{Gal}(Y/X)$, by the Galois correspondence, there exist corresponding subcoverings of Y/X . Denote by Z_{K_i} the Fox completion of the covering corresponding to G_{K_i} and by T_{K_i} the Fox completion of the covering corresponding to I_{K_i} ; these are the *decomposition* and *inertia spaces*, respectively.

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Theorem: “Hilbert”[Mor12]

The map $M \rightarrow T_{K_i}$ is a ramified covering of degree $e := |I_{K_i}|$ ramified over the image of K_i (say $K_{i,T}$) with ramification index e . The map $T_{K_i} \rightarrow Z_{K_i}$ is a cyclic covering of degree $f := |\text{Gal}(K_i/K)| = |G_{K_i}/I_{K_i}|$ such that the covering degree of $K_{i,T}$ over $K_{i,Z}$ is f . The map $Z_{K_i} \rightarrow S^3$ is a covering of degree r such that K is completely decomposed into an r -component link containing $K_{i,Z}$ as a component.

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